

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A
Mock Interview

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

- 27. How would you present your simulation results to a client? (BL4)(1)
- 28. Explain jitter using a real-world analogy (BL5)(1)
- 29. Describe how you tested network performance in a lab (BL4)(1)
- 30. Explain the importance of performance metrics in one structured answer (BL5)(1)
- 31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
- 32. Why are you interested in networking and simulation tools? (BL5)(1)
- 33. What makes you confident in analyzing network performance? (BL5)(1)
- 34. Explain a concept you are very strong in (related to networking) (BL5)(1)
- 35. How would you handle a question you don't know? (BL5)(1)
- 36. Describe a situation where you solved a technical problem (BL4)(1)
- 37. Why should we select you for a networking role? (BL5)(1)
- 38. What challenges did you face while learning Packet Tracer? (BL5)(1)
- 39. How do you improve your technical skills regularly? (BL5)(1)
- 40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. How does network congestion affect packet delay and packet loss?

Network congestion occurs when traffic exceeds the network capacity. Packets wait in queues, increasing delay. When buffers become full, packets are dropped, causing packet loss and retransmissions.

2. Compare star and mesh topologies on reliability and fault tolerance.

A star topology is easy to manage, but failure of the central switch can affect the whole network. A mesh topology has multiple paths, so if one link fails, traffic can use another path. Therefore, mesh provides higher reliability and fault tolerance but costs more.

3. Why does increasing the number of routers increase end-to-end delay?

Each router introduces processing and queuing delay and may also add transmission delay. Therefore, as the number of routers increases, the total end-to-end delay generally increases.

4. How does QoS improve voice and video traffic?

QoS prioritizes real-time traffic such as voice and video over less-sensitive traffic. It reduces delay, jitter, and packet loss, resulting in clearer calls and smoother video.

5. Impact of fiber optic instead of copper cables.

Fiber provides higher bandwidth, lower signal loss, longer transmission distances, and better resistance to electromagnetic interference. However, fiber installation and equipment can initially cost more than copper.

6. How does VLAN segmentation improve performance and security?

VLANs divide a network into logical segments. This reduces unnecessary broadcast traffic and improves performance. It also isolates departments or sensitive systems, improving security and access control.

7. TCP vs UDP impact on performance.

TCP provides reliable, ordered delivery using acknowledgments and retransmissions, but this creates overhead. UDP has lower overhead and latency but does not guarantee delivery. TCP suits web browsing and file transfers, while UDP is suitable for VoIP, gaming, and streaming.

8. Why is load balancing important in enterprise networks?

Load balancing distributes traffic across multiple servers or links. It prevents individual resources from becoming overloaded, improving performance, availability, scalability, and fault tolerance.

9. How do frequent routing updates affect network efficiency?

Routing updates consume bandwidth and router processing resources. Excessive updates can increase overhead and reduce efficiency. However, timely updates are necessary to maintain accurate routes when network conditions change.

10. Trade-off between security and network performance.

Firewalls, encryption, and intrusion detection improve security but require processing resources and may introduce some latency. Properly configured hardware and selective security policies can provide strong protection while minimizing performance impact.

11. A network has high delay—how will you identify and fix it?

I would use ping and traceroute to identify where delay occurs. Then I would check

bandwidth utilization, router CPU usage, queues, cables, and routing paths. I would reduce congestion, optimize routing, or upgrade the affected link.

12. Low throughput despite high bandwidth—why?

Possible reasons include packet loss, network congestion, high latency, inefficient protocols, hardware limitations, interference, server limitations, or faulty cables.

13. How would you reduce packet loss in a congested network?

I would identify the congested link, apply QoS, prioritize important traffic, use load balancing, increase bandwidth if necessary, and optimize routing. I would also check for faulty network equipment.

14. VoIP call is breaking frequently—how troubleshoot?

I would check packet loss, latency, jitter, bandwidth utilization, and connectivity. Then I would verify QoS configuration, router interfaces, cables, and VoIP device settings. I would use ping and network monitoring tools to locate the problem.

15. Network has high jitter—suggest improvements.

I would apply QoS to prioritize real-time traffic, reduce congestion, use stable routes, increase available bandwidth, and check for overloaded or unreliable links.

16. How will you simulate congestion in Packet Tracer?

I would create multiple devices and generate traffic through a common link. Then I would limit the link bandwidth or increase traffic volume and observe delay, packet loss, and throughput using simulation and ping tools.

17. Packets are delayed in a router—what steps will you take?

I would check router CPU and memory utilization, interface errors, queue size, bandwidth utilization, routing configuration, and packet drops. I would then optimize routing, apply QoS, or upgrade the affected interface.

18. How would you optimize a network for online gaming?

I would prioritize gaming traffic using QoS, minimize latency and jitter, reduce packet loss, use efficient routing, avoid congestion, and use a stable high-speed connection.

19. Design a small network and explain how you measure performance.

I would connect several PCs to a switch and router and configure IP addresses. I would measure throughput, latency, packet loss, jitter, and bandwidth utilization using ping, simulation tools, and network monitoring.

20. If RTT is very high, how diagnose it?

I would use ping and traceroute to identify the problematic hop. Then I would check congestion, routing paths, link quality, bandwidth utilization, and router performance. Finally, I would optimize the route or replace/upgrade the problematic link.

21. Explain network performance to a non-technical person.

Network performance means how well and how quickly a network delivers information. Just like a road, a good network allows data to travel quickly without traffic jams or lost vehicles.

22. Describe throughput using a real-life example.

Throughput is the actual amount of data successfully delivered per second. For

example, if a water pipe delivers 100 liters per minute, that delivery rate is similar to network throughput.

23. How does packet loss affect video streaming?

Packet loss means some video data does not reach the destination. This can cause buffering, frozen frames, reduced quality, or interruptions in the video.

24. Explain delay types in under 1 minute.

There are four main delays: processing delay for examining a packet, queuing delay while waiting in a router, transmission delay for putting the packet onto the link, and propagation delay for the signal to travel through the medium.

25. Describe Cisco Packet Tracer professionally.

Cisco Packet Tracer is a network simulation and learning **tool** used to design, configure, and troubleshoot virtual network environments. It allows users to test devices, protocols, routing, and network behavior without requiring physical hardware.

26. Explain QoS simply and technically.

Simply, QoS is like giving an emergency vehicle priority on a busy road. Technically, QoS classifies and prioritizes network traffic so delay-sensitive applications such as VoIP and video receive appropriate bandwidth and lower delay.

27. How would you present simulation results to a client?

I would present the network design, test conditions, key metrics, graphs/tables, problems identified, and recommended improvements. I would focus on business impact rather than unnecessary technical details.

28. Explain jitter using a real-world analogy.

Jitter is variation in packet arrival time. Imagine buses that should arrive every 5 minutes, but one arrives after 2 minutes and the next after 10 minutes. The irregular arrival represents jitter.

29. Describe how you tested network performance in a lab.

I configured the network, generated traffic, and measured latency, throughput, packet loss, and jitter. I used tools such as ping and Cisco Packet Tracer simulation to analyze the results.

30. Explain the importance of performance metrics.

Network performance metrics help identify problems and evaluate network quality. Throughput shows data delivery rate, latency shows delay, packet loss shows reliability, and jitter shows variation in delay. Together, they help administrators optimize the network.

31. Introduce yourself and explain your knowledge of network performance.

I am a Computer Science Engineering student with an interest in networking. I have learned concepts such as bandwidth, throughput, latency, packet loss, jitter, QoS, routing, and network topologies. I have also used Cisco Packet Tracer to design and analyze networks.

32. Why are you interested in networking and simulation tools?

I am interested in networking because it connects systems and enables communication.

Simulation tools like Cisco Packet Tracer allow me to practice network configuration, troubleshooting, and performance analysis safely before working with real equipment.

33. What makes you confident in analyzing network performance?

My understanding of performance metrics and systematic troubleshooting gives me confidence. I analyze problems using measurable values such as latency, packet loss, throughput, and jitter rather than making assumptions.

34. Explain a concept you are very strong in.

I am strong in QoS. QoS manages network traffic by assigning priority to important applications. For example, voice and video can be prioritized over less-critical downloads, reducing delay and improving communication quality.

35. How would you handle a question you don't know?

I would be honest that I don't know the answer. I would explain what I do know, reason through the problem logically, and then learn or verify the correct solution.

36. Describe a situation where you solved a technical problem.

While working with a network simulation, if communication failed, I would systematically check the IP addresses, subnet masks, interfaces, cables, and routing configuration. By testing each layer step by step, I can identify and correct the configuration problem.

37. Why should we select you for a networking role?

You should select me because I have a strong foundation in networking concepts, a willingness to learn, and a systematic approach to troubleshooting. I can understand technical problems and explain solutions clearly.

38. What challenges did you face while learning Packet Tracer?

Initially, understanding device configuration, IP addressing, routing, and troubleshooting was challenging. I overcame this through repeated simulations and step-by-step testing, which improved my confidence.

39. How do you improve your technical skills regularly?

I regularly practice programming and networking concepts, perform simulations, solve technical problems, and learn from documentation and practical examples. I also try to apply theoretical concepts through hands-on projects.

40. Demonstrate confidence while explaining a complex concept simply.

For example, routing can be explained simply as choosing the best road for data to reach its destination. Routers examine the destination address and select an appropriate path. In a complex network, routing protocols automatically exchange information and help routers choose efficient paths. This makes communication reliable and efficient.